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SUMMARY

Robotics and controls engineer, 4+ years taking safety-critical medical automation from R&D to production. Real-time control at 1 kHz, signal processing, computer vision and embedded C++ on regulated hardware. MS Robotics and Controls, Columbia (ROAM Lab).

EXPERIENCE

Clarapath Inc. — Robotics and Controls Engineer

Hawthorne, NY — Apr 2022 – Present

Controls technical lead, 6-person controls and AI group, mentoring and onboarding new hires and interns, collaborated with pathologists/histologists. Product: a robotic dual microtome system that slices tissue microns thin onto slides using gantries and a tape system (web handler).

- **Built** trajectory optimization for microtome paths using control barrier functions, nonlinear optimization and Bezier curves, under safety constraints - process time 2x faster (1.6s -> 0.8s) [MATLAB, Python, C++]
- **Introduced** cut-signal processing for time/block invariance using PCHIP resampling and Welford statistics, real-time error detection and feature extraction; peak detection on undersampled signals to measure section and block characteristics for user and consumable safe operation [C++, Python]
- **Led** a cross-functional root-cause investigation on section quality; conducted multiple design-of-experiments to isolate root cause, then the mechanical redesign and control rework to run without it — defect rate 19% to under 1% [MATLAB, Python]
- **Built** the product's data-analysis capability from scratch: contact detection and closed-loop depth estimation at the blade to refine block-level cutting; segmentation of block/tissue regions [Python, OpenCV, PyTorch]
- **Owned** controls and embedded software for 10+ subsystems on a multi-node ROS architecture — real-time signal processing, 1 kHz adaptive control, finite state machines, and 0.25 mm laser landmark calibration (plane-normal fit, Rodrigues rotation to CAD nominal) and motion planning for 3-DOF gantries, microtome and tape driver - reduced production time and cost by 3x [C++, SQL, Python]
- **Designed** the machine-to-server telemetry stack for continuous data logging/plotting from fielded units with safe file-handling [C++, SQL]
- **Built** power-on self-test sequences, firmware versioning through CI/CD and acceptance test protocols to statistically validate builds on the manufacturing floor and ensuring traceability [Python, SQL]
- **Introduced** user-independent inter-subsystem processes, e.g. block-to-tape pressing, with closed-loop control, image analysis and calibration processes removing build variation tuning and improving deliverable quality [C++, Python]
- **Designed** the error-handling architecture of multi-layer error codes and loggers to a <1% field error rate; introduced dependency-ordered safe-mode entry and initialization; raised machine throughput 1.6x [C++, Bash, Python]
- **Enforced** analytical sizing and tolerance analysis before fabrication instead of build-and-test; rigorous documentation for code reviews — 2x fewer design iteration cycles
- **Devised** innovative solutions by means of linearized nonlinear controls, Fourier analysis, grid search and signal filters enabling machine critical functionality while reducing mechanical redesign costs [MATLAB, Python, C++]

Columbia University, ROAM Lab (Dr. Matei Ciocarlie) — Graduate Research Assistant

New York, NY — Nov 2020 – Jan 2022

- **Built** controls firmware for a modular biomimetic serpent robot, applying model-free reinforcement learning policies, model predictive control and PID loops to locomotion and state estimation [Python, C++, ROS]
- **Integrated** IMUs, magnetic encoders and optical sensors over a CAN network at 100 Hz, logging into ROS nodes [C++]

Evtek — Machine Learning Research Intern

New York, NY — Sep 2021 – Dec 2021

- **Deployed** deep learning models for identification, masking and sorting in automated recycling, and improved conveyor object recognition and tracking 40% using Kalman filtering [OpenCV, PyTorch]

EDUCATION

Columbia University, School of Engineering and Applied Science

New York, NY — Dec 2021

MS Mechanical Engineering (Robotics and Controls), GPA 3.8. TA: Controlled Dynamic Systems, ML for Mechanical Systems.

New York University, Tandon School of Engineering

Brooklyn, NY — May 2020

BS Mechanical Engineering, cum laude. GPA 3.6. Minors in Aerospace Engineering, Nuclear Engineering and Computer Science.

SELECTED PROJECTS

- **Robotics algorithms** (github.com/geojob): RRT and MoveIt motion planning, Cartesian control and inverse kinematics on UR5 and KUKA arms, particle-filter state estimation, model-free RL and evolutionary algorithms [C++, Python, Julia, ROS]

SKILLS

Languages: C++, Python, MATLAB, SQL, Bash, Java, Julia, CUDA, R

Control & Estimation: Model Predictive Control, real-time control, adaptive control, optimal control, nonlinear control, control barrier functions, state estimation, time series analysis, stochastic modeling

Robotics: ROS, motion planning, manipulation, inverse kinematics, sensor fusion

Tools & Simulation: Gazebo, MuJoCo

Machine Learning and Vision: PyTorch, TensorFlow, JAX, OpenCV, deep learning, reinforcement learning, imitation learning, image segmentation, object tracking

Software and Design: Git, version control, continuous integration, unit test harnesses, Linux, Docker, embedded firmware, SOLIDWORKS, CATIA, ANSYS, Altair